PAH-OMC-018: Unaccelerated generator limit and stationary local pre-form

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

first issued 2026-09-06 · this version issued 2026-09-06 · v1.0

Purpose and scope

This checkpoint tests the original PAH generator on the resolved radial path, not a new Hamiltonian, rate law or physical sector. It uses precisely the PAH-001 functional and PAH-OMC-001 directed root labels on the OMC-004 anchored two-row strip G_n, $n \geq 2$, including its unsplit terminal square. The source parameters are $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$ and the remaining displayed couplings one. The volume is $|V_n| = 2(n + 2)$; this relational strip has no identified physical volume, dimension or length. First take

$$h_j = 2^{-j}, \quad R_j = 2^j, \quad M_j = 2^{2j}, \quad r_v = h_j \ell_v, \qquad j \longrightarrow \infty,$$

at fixed n, then take $n \to \infty$. There is no time acceleration. The all-Q state has exactly $w_Q = Z_Q/Z$ and labelled counting weights, retaining phase labels at zero amplitude. R-509 gives the fixed-n limit $\nu_n \propto e^{-F_n} \prod_v dr_v$ times label counting. R-510 supplies the specified anchored common local state ν_∞. No radial Jacobian, gauge quotient, counterterm, optional probe or boundary value is added.

The source preregistration is immutable and precedes calculations:

```
strategy/pa-hyp/PAH-OMC-018-generator-prereg-v1.json
92a4fa92cb8c0384812220acd2e9929539a10a114fe348ea3f8aa7676f98aa05
```

Its parent pins include the final R-510 publication manifest, not the historical prepublication hash. Detailed proofs and applicability crosswalk are in the companion generator certificate. This note is their one synthesis.

Frozen domain and statement

Let $\mathcal{D}$ be the real gauge-invariant bounded cylinder algebra that is globally Lipschitz in amplitude in the ℓ^1 norm, uniformly over labels. Write $M_f = \|f\|_\infty$ and choose L_f with

$$|f(r, z) - f(r', z)| \leq L_f \sum_v |r_v - r'_v|.$$

For support in the R-510 prefix Λ_m, set $N(f) = \max(2, m + 2)$. The original root-rate closures stabilize by this stage. Let H_f count potential directed radial roots incident to read amplitudes, and D_f count potential phase, link and aperture roots that can change f. In terms of read vertices V_f and links E_f,

$$H_f \leq 2d_{\max}|V_f|, \quad d_{\max} = 5, \qquad D_f \leq 4|V_f| + 2|E_f|.$$

Coincident $K = 2$ phase and link maps remain separate directed channels. The sampling map $S_{n,j}$ evaluates the same amplitude function on the grid; it is not a Gibbs pushforward or an isometry between Hilbert spaces.

The derived limiting local operator A_n consists of the original phase, link and aperture summands at continuous amplitudes. For every $n \ge N(f)$,

$$\begin{aligned} \|L_{n,j}S_{n,j}f - S_{n,j}A_n f\|_{L^2(\mu_{n,j})} &\le 2H_f L_f h_j, \\ \mathcal{E}_{\mathrm{TR},n,j}(S_{n,j}f, S_{n,j}f) &\le 2H_f L_f^2 h_j^2. \end{aligned}$$

The estimate is uniform in j and n for each fixed support. It is not a global norm bound on the full generator. The subsequent ordered passage gives a dense local pre-form

$$\nu_\infty(Ag) = 0, \qquad \mathcal{E}_\infty(f, g) = -\nu_\infty(fAg) = \frac{1}{2}\sum_r \nu_\infty[c_r \Delta_r f \Delta_r g].$$

It is symmetric and nonnegative. Density of the test algebra is not graph-core density, closability or construction of an infinite-volume process.

Inverse transport: the uniform estimate

For an admissible root write $y = T_r x$. The exact rates give

$$c_r(x) = m_r(x)e^{-(F(y)-F(x))/2}, \quad \pi(x)c_r(x) = m_r(x)\sqrt{\pi(x)\pi(y)}, \quad \pi(x)c_r(x)^2 = m_r(x)^2\pi(y).$$

Since $0 < m_r \le 1$, inverse transport on its exact partial domain gives $\int \pi c_r^2 \le 1$, and Cauchy–Schwarz gives $\int \pi c_r \le 1$. The radial donor and receiver constraints are both retained; the inverse image is a subset, not falsely the whole state space. No cutoff-independent pointwise upper bound for c_r is used. Every radial jump has amplitude ℓ^1 distance $2h_j$, hence $|\Delta_r f| \le 2L_f h_j$. Finite-sum Cauchy–Schwarz proves the generator bound; the directed factor $1/2$ proves the form bound. No paired-direction cancellation is assumed, so lower and upper endpoints are included.

Nonradial actions and rates at sampled amplitudes agree exactly with A_n. Therefore $L_{n,j}S_{n,j}f - S_{n,j}A_n f = L_{\mathrm{TR},n,j}S_{n,j}f$. On both discrete states and ν_n, the same square identity implies $\|A_n f\|_2 \le 2D_f M_f$. The function $A_n f$ may be unbounded and outside $\mathcal{D}$. A shrinking radial root cannot be identified with the old integer-coordinate root under a dyadic embedding.

The two limit passages

For fixed n, use the exact R-509 half-open cell representation, including the upper grid endpoint. Put $a_j(t) = h_j\lfloor t/h_j \rfloor$. A two-root term of $e^{-F}(A_n f)^2$ carries the weight

$$m_r m_s \exp\left[-\frac{F(T_r x) + F(T_s x)}{2}\right].$$

Nonradial moves retain all amplitudes. Each source energy is bounded below by $\sum_v r_v^6/6$. Thus the sampled terms, with bounded increments, are dominated at fixed n by a constant times

$$\prod_v \exp[-\max(t_v - 1, 0)^6/6].$$

This integrable envelope and the positive limiting partition function justify the mesh passage of pairings, forms and second moments. Radial cross terms vanish by the proved form bound. Detailed balance then gives $\nu_n(A_n g) = 0$ and the symmetric directed local form at fixed n.

For the n passage, the rate of a root acting on f uses only a finite interaction closure. Phase and aperture roots at column at most m use edges and faces reaching at most $m+1$; radial neighbors can reach

$m+2$. For $n \geq N(f)$ all relevant split cells and edges already exist. The terminal square is preserved in the state but lies outside this rate increment. Hence $A_n f$ is the same local function Af eventually.

Let $B_f = 2D_f M_f$. R-510 prefix total-variation convergence applied to $\min((Af)^2, K)$ gives $\nu_\infty((Af)^2) \leq B_f^2$. For ν_n and ν_∞ alike,

$$\int |Af| \, \mathbf{1}_{\{|Af|>K\}} \leq B_f^2/K.$$

Truncate, pass the bounded local part, then remove the tail. This proves the stationary pairings and pre-form limit. The same argument applies to single conductances, whose second moments are at most one. Bounded weak convergence alone would not license this unbounded-generator passage. The order remains j first, n second.

The activity discriminator

Every amplitude-only $f \in \mathcal{D}$ has $Af = 0$ and $\mathcal{E}_\infty(f,f) = 0$. Nevertheless $b_v = \min(1, r_v)$ has positive variance under the positive R-510 prefix density. Static nondegeneracy does not imply radial motion. In contrast the aperture observable s_v has positive form energy, since its admissible aperture move has positive rate and increment $1/2$. Thus the limiting local operator is not zero: it retains label jumps and freezes amplitudes at the original time scale. This is a bounded kinetic limitation, not a refutation of the full PAH model.

Devil's-advocate audit and verification boundary

The primary lane uses squared source edges; the non-importing independent lane uses expanded signed edges and square-root Gibbs weights. Exact root fixtures, mobility factors and interaction closures agree. Hostile checks attack the half exponent, Gibbs sign, directed factor, duplicate channels, partial endpoints, false radial embedding, absent uniform integrability and improper deletion of global energy. Eleven Lean declarations check the algebraic weight identities, finite-sum bound, coefficients, tail inequality and activity arithmetic. Integral limits and operator domains remain an analytic proof, not a fully Lean-formalized theorem.

- Wrong exponent or factor: DISMISSED by independent identities and mutation tests; the wrong conventions are explicitly rejected.

- Ignored endpoint or duplicate root: DISMISSED by exact inverse subsets and label-distinct enumeration, without rate fitting.

- Divergent pointwise rates: UPHELD as a failed shortcut; the proof uses weighted inverse transport, not a pointwise envelope.

- Weak convergence alone passes Af: UPHELD against that shortcut; the sextic envelope and second-moment tail bound supply the missing step.

- Dense pre-form means dynamics: UPHELD. Closure and process/semigroup convergence remain separate gates.

Background convention: Aldous–Fill, chapter 3, section 3.6.1, equations (3.74)–(3.76), continuous-time directed Dirichlet form. The finite identity is rederived; no external infinite-process theorem is imported. This is standard reversible analysis applied to this exact PAH path, not a claim of a new general mathematical method. External review of the inverse-domain and unbounded-rate passages is invited.

Result footer

```
Result ID: R-511
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: Sampled original-generator j-limit and ordered
 stationary symmetric nonnegative local pre-form; radial part vanishes.
Scope: PAH-001 / OMC-001 / OMC-004 / OMC-016 fixed parameters;
 j first, then n; bounded Lipschitz amplitude cylinders.
Dependencies: Frozen source packets, R-509 and R-510 exact state inputs.
Evidence grade: ANALYTIC, EXACT, EXECUTED, INHERITED state inputs.
Reproduction command: python -X utf8
 verification/scripts/pah_omc018_verify.py --check
Expected output: PAH-OMC-018 INTEGRATED: PASS; Lean PASS.
Falsification gate: Source drift, inverse-domain/factor error,
 false interaction closure, or failure of the weighted tail passage.
Tier before / after: T1 / T1 (host unchanged); auxiliary_support.
No-overclaim statement: No closed form, infinite-volume dynamics,
 physical Pre-A, spacetime, QFT, gravity, continuum or TOE result.
Next required action: Test closability of the frozen local pre-form
 and preservation of its radial nullspace in its minimal closure.
```